

# Isaac Weintraub

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## Personal Statement

Mission-driven leader and technical expert specializing in control theory, optimization, and tactical autonomy in the area of aerospace systems. Proven ability to guide strategic research initiatives, manage multi-million dollar resources, and lead high-performing international teams to deliver advanced tactical autonomy solutions for the warfighter.

## Employment

### Air Force Research Laboratory

Dayton, OH

Autonomous Controls Branch

29 Mar 2021 - Present

Electronics Engineer, DR-3 (40hr/week)

- Managed multi-\$M R&D portfolio in multi-agent control and coordination partnering with academic, commercial, and government partners. Collaborated with these partners to publish and co-author technical contributions for the aerospace industry in globally attended conferences and venues.
- Performed research in controls for defense applications, specifically for air-breathing systems, leveraging differential game theory, optimal control theory, and other techniques to inform beyond visual range tactics
- Supported multi-agent unmanned aerial vehicle flight tests for multiple missions: cooperative convoy escort, automatic simultaneous arrival, automatic persistent ISR, automatic engagement zone avoidance, automated aircraft survivability, and run-time assured autonomy
- Developed relational maneuvering methods and tested them in flight simulators located in the aerospace vehicles technology assessment and simulations facility.

Aerospace Vehicles Technology Assessment and Simulations Branch

29 Sep 2015 - 26 Mar 2021

Electronics Engineer, DR-2 (40hr/week)

- Performed system identification and implemented an inner-loop controller on an FPGA for the Large Amplitude Multimode Aircraft Simulator (LAMARS) using LabVIEW
- Led motion tuning and flying/handling qualities development for classified research programs held in the LAMARS motion simulator. Designed control systems in MATLAB Simulink for LAMARS.
- Lead IADS test and evaluation for Auto GCAS at Edwards AFB for 4 F-16 flights
- Published and reviewed journal and conference publications for the IEEE and AIAA
- Managed the manufacturing facility in Building 145; supported 10 separate AFRL branches, saving over \$3M in manufacturing costs while securing the intellectual property of the USAF
- Managed Missile DATCOM for the USAF; supported 7 SBIRs, 28 Commercial Companies, and the 5I's Nations for developing future weapon systems

Air Force Research Laboratory - Maker Hub

17 Jan 2017 - 21 Apr 2017

Director of Operations, Sabbatical (40hr/week)

- Trained S&E's from AFRL, NASIC, 711th, and other DoD facilities on how to go from napkin concepts to functional prototypes using CAD/CAM, additive and subtractive manufacturing, PCB design, programming, and the design iteration process
- Managed, developed, and marketed a rapid prototyping facility supporting cross-DoD facilities

**Infoscitex Corporation****Dayton, OH**

Research Scientist (40hr/week)

05 Jan 2015 - 29 Sep 2015

- Designed and fabricated actuator hardware in the loop testing station for thermal/energy applications, resulting in an AFRL patent
- Manufactured and tested ultra-lightweight flapping wing micro air vehicles, resulting in the first flight of a minimally actuated micro air vehicle with only two actuators
- Wrote proposals for flexible air vehicles and thermal management research, resulting in a funded task order (RAST)

**General Dynamics Information Technology****Dayton, OH**

Scientist (40hr/week)

02 Jan 2012 - 02 Jan 2015

- Supported flapping wing micro air vehicle research for Air Force Research Laboratory, resulting in an AFRL patent, journal and conference papers, book chapters, and technical reports
- Developed test and evaluation equipment using high-speed cameras and sensors
- Designed printed circuit boards for control of flapping wing micro-aerial vehicles
- Fabricated flapping wing micro-aerial vehicles using non-traditional methods of manufacturing

**Universal Technology Corporation****Dayton, OH**

Summer Research Fellow (40hr/week)

06 May 2011 - 20 Dec 2011

- Designed, manufactured, and tested flapping wing micro-aerial vehicles and conducted bench-level tests resulting in successful proof of concept air table experiments
- Demonstrated the suitability and efficacy of split cycle control laws for flapping-wing flight with hardware-in-the-loop tests

**Automation and Robotics Research Institute****Fort Worth, TX**

Graduate Research Assistant (40hr/week)

03 Aug 2009 - 27 May 2011

- Designed and machined parts for DARPA funded confidential project partnering with Lockheed Martin
- Researched and presented micro-robotic locomotion modules for defense applications
- Lectured, graded work, held lab sessions, and provided regular office hours for the Microprocessor Systems undergraduate course
- Opened and provided tutoring for any engineering student during regularly scheduled office hours

**TolTest Incorporated****Maumee, OH**

Engineering Intern (40hr/week)

06 Jun 2006 - 26 Sep 2005

- Worked and designed government and commercial products for a global market
- Produced civil, mechanical, and electrical CAD Models using AutoCAD

**Toledo Molding and Die****Toledo, OH**

Engineering Intern (40hr/week)

06 Jun 2006 - 26 Sep 2005

- Designed and constructed testing equipment for performance and durability testing and evaluation of prototype automotive parts
- Analyzed and reported technical data and specifications to automotive customers

## Awards / Nominations

### **Outstanding Defense of Department Civilian Award (Department of War)**

**Award 2026**

For exemplary service and for dedication to executing the transformational changes necessary to revive the warrior ethos, rebuild our (the United States of America's) military, and reestablish deterrence.

### **Don Ross Award (AFRL/RQ)**

**Award 2026**

As a member of the Cooperative Control Team, we were recognized for developing and transitioning foundational autonomy skills that directly enabled the fielding of Autonomous Collaborative Platforms (ACPs) for offensive and defensive counter-air missions, providing the DoD with a decisive advantage in contested, near-peer environments.

### **Civilian Category III (Non-supervisory) of the Year (AFRL/RQQ)**

**Nomination 2025**

### **Team Quarterly (Q4) Award (AFRL/RQQ)**

**Award 2025**

### **International Team Award (AFRL)**

**Award 2025**

The team successfully integrated advanced autonomy technologies through a collaborative project with Australia, the United Kingdom, and the United States, to improve mission planning capabilities for the multinational alliance.

### **Dr. Richard B. Rivir Scientific and Technical Achievement (AFRL/RQ)**

**Nomination 2023 & 2024**

A notable and distinguished in-house technical achievement. "For contributions in the area of tactical autonomy."

### **Gen. Benjamin D. Foulois Award (AFRL/RQ)**

**Award 2022**

For significant and outstanding in-house science resulting in substantial, visible, tangible evidence of a new or improved product, capability or service. Factors considered include significant value added to the engineering body of knowledge or the capabilities of the directorate, engineering excellence, originality, creativity, ingenuity, quality of workmanship and importance to the Air Force. This award recognizes a culmination, multi-year outstanding achievement.

### **Dr. Courtland D. Perkins Award (AFRL/RQQ)**

**Nomination 2021**

For significant and outstanding in-house engineering resulting in substantial, visible, tangible evidence of a new or improved product, capability or service. Factors considered include significant value added to the engineering body of knowledge or the capabilities of the directorate, engineering excellence, originality, creativity, ingenuity, quality of workmanship and importance to the Air Force. This award recognizes a culmination, multi-year outstanding achievement.

### **Director's Trophy - Classified Category (AFRL/RQ)**

**Nomination 2019 / Award 2020**

For significant contributions to the RQ mission of AFRL that had a significant impact and enhanced the credibility of AFRL. This award recognizes significant contributions that can otherwise not be recognized due to its sensitivity.

### **Collier Trophy, Automatic Ground Collision Avoidance System Team**

**Award 2018**

Part of the Auto GCAS team as a flight test engineer at Edwards AFB, The Robert J. Collier Trophy is an annual aviation award administered by the U.S. National Aeronautic Association (NAA), presented to those who have made "the greatest achievement in aeronautics or astronautics in America, with respect to improving the performance, efficiency, and safety of air or space vehicles, the value of which has been thoroughly demonstrated by actual use during the preceding year."

Role: Flight Test Support and Program Management Support

### **Best Technical Presentation, 40th AIAA DCASS**

**Award 2015**

### **Best Technical Presentation, 39th AIAA DCASS**

**Award 2014**

### **Judges Volunteer Award, FIRST Tech Challenge**

**Award 2014**

### **Atmospheric Flight Mechanics Best Paper, AIAA**

**Award 2013**

|                                                                                 |                        |
|---------------------------------------------------------------------------------|------------------------|
| <b>Local Compass Award, FIRST Tech Challenge</b>                                | <b>Award 2011</b>      |
| <b>State Compass Award, FIRST Tech Challenge</b>                                | <b>Award 2011</b>      |
| <b>Bronze Presidential Volunteer Service Award</b>                              | <b>Award 2011</b>      |
| <b>Jack Fitzler Endowed Scholarship Award, University of Texas at Arlington</b> | <b>Award 2011</b>      |
| <b>Best Student Award, Automation and Robotics Research Institute</b>           | <b>Award 2010</b>      |
| <b>Barry M. Goldwater Nominee, Rose-Hulman Institute of Technology</b>          | <b>Nomination 2008</b> |

## Mentorship and Program Sponsorship

Total Funding \$215k

|                                                            |      |
|------------------------------------------------------------|------|
| • Alexander Denton, Air Force Institute of Technology, PhD | 2027 |
| • Jordan Graham, Air Force Institute of Technology, MS     | 2026 |
| • Drew Scott, University of Cincinnati, PhD                | 2024 |
| • Alex Ladadio, Air Force Institute of Technology, MS      | 2024 |
| • Carl Gotwald, Air Force Institute of Technology, PhD     | 2024 |
| • Justin Merrick, Air Force Institute of Technology, PhD   | 2024 |
| • Patrick Dillon, Air Force Institute of Technology, MEng  | 2022 |
| • Matthew Jones, Air Force Institute of Technology, MEng   | 2022 |
| • Shawn Stephens, Air Force Institute of Technology, PhD   | 2021 |

## Education

|                                                                                                     |                   |
|-----------------------------------------------------------------------------------------------------|-------------------|
| <b>Air Force Institute of Technology</b>                                                            | <b>Dayton, OH</b> |
| Doctor of Philosophy in Electrical Engineering, GPA 3.9/4.0                                         | 26 Mar 2021       |
| Dissertation Title: Optimal Defense of High Value Airborne Assets                                   |                   |
| Committee: Meir Pachter (chair), Richard Cobb, William Baker, Lt. Col. Michael Zollars, Eloy Garcia |                   |

|                                                           |                      |
|-----------------------------------------------------------|----------------------|
| <b>University of Texas at Arlington</b>                   | <b>Arlington, TX</b> |
| Masters of Science in Electrical Engineering, GPA 3.8/4.0 | 13 Aug 2011          |

|                                                                              |                        |
|------------------------------------------------------------------------------|------------------------|
| <b>Rose-Hulman Institute of Technology</b>                                   | <b>Terre Haute, IN</b> |
| Bachelor of Science (Magna Cum Laude) in Mechanical Engineering, GPA 3.7/4.0 | 30 May 2009            |
| Minor in Electrical Engineering, Certificate in Robotics                     |                        |

## Skills

### Hands on Skills

Sheet Metal Fabrication, Carbon Fiber and Fiberglass Molding and Fabrication, Mill, Lathe, Welding (MIG/Stick), Circuit Design and Debugging, CNC Mill, CNC Lathe, Laser Cutting, 3D Printing (SLA FDM SLS), PCB Design and Fabrication, Woodwork, Power Tools and Hand Tools, Automotive Restoration / Fabrication

### Technical Skills

Altium, Autodesk Inventor, AutoCAD, Eagle CAD, MasterCAM, Solid Edge, Solid Works, Labview, Labview Real-Time, MATLAB, Simulink, Maple, Microsoft Office, Open Office, Google Docs, PCB Design, Visi-Peps Wire, Adobe Photoshop, Basic, C, C#, G-Code, LaTeX, TikZ, Typst

## Museum Pieces

- FWMAV-1 - United States Air Force Museum. Bench-level air vehicle used for initial prototyping and testing for flapping wing micro air vehicle wing testing and measurement.
- FWMAV-2 - United States Air Force Museum. Air-Table prototype vehicle used to prove the efficacy and suitability of split-cycle control laws for flapping-wing flight.
- FWMAV-3 - United States Air Force Museum. Airworthy flight vehicle prototype capable of flying controllably in the air with only two actuators.

## Publications

### Dissertation

- [1] **Isaac Weintraub** "Optimal Defense of High Value Airborne Assets", Dissertation, Air Force Institute of Technology, 2021. [html: https://scholar.afit.edu/etd/4534/](https://scholar.afit.edu/etd/4534/).

### Patents and Disclosures

- [1] **Isaac Weintraub**, Alexander Von Moll, "Controlled Patent", AFRL Patent Disclosure, 2024.
- [2] **Isaac Weintraub**, Jason Shroyer, Nathan Ritsema, Dillon Sluss, "Synthetic Waypoint Guidance using Bezier Paths", AFRL Patent Disclosure, 2020.
- [3] Michael W. Oppenheimer, **Isaac Weintraub**, David O. Sigthorsson, David B. Doman, "Methods and Apparatus to Achieve Independent Six Degree Control of Flapping Wing Micro Air Vehicle", Awarded US Patent 9,428,269 Aug 2015.

### Book Chapter

- [1] **Isaac Weintraub**, David O. Sigthorsson, Michael W. Oppenheimer, David B. Doman, "Implementation of Split-Cycle Control for Micro Aerial Vehicles", In J. H. Kim, E. Matson, H. Myung, P. Xu, F. Karray (Eds.) Robot Intelligence Technology and Applications 2 (pp 859-876) Springer (2014). doi: 10.1007/978-3-319-05582-4.

### Journal

- [1] Prajakta Surve; Shaunak D. Bopardikar; Alexander Von Moll; **Isaac Weintraub**; David W. Casbeer, "Heterogeneous Pursuit of an Active Target Under Sensing Constraints", in IEEE Control Systems Letters, vol. 9, pp 3017-3022, 2025, doi: 10.1109/LCSYS.2025.3645033
- [2] Thomas Chapman, Alexander Von Moll and **Isaac Weintraub**, "Safe Navigation in the Presence of Range-Limited Pursuers," in IEEE Control Systems Letters, vol. 9, pp. 2849-2854, 2025, doi: 10.1109/LCSYS.2025.3645221.
- [3] Prajakta Surve, Shaunak D. Bopardikar, Alexander Von Moll, **Isaac E. Weintraub** and David W. Casbeer, "Mutual Support by Sensor-Attacker Team for a Passive Target," in IEEE Open Journal of Control Systems, vol. 4, pp. 418-432, 2025, doi: 10.1109/OJCSYS.2025.3612246.
- [4] Dejan Milutinović, Alexander Von Moll, Satyanarayana G. Manyam, David W. Casbeer, **Isaac E. Weintraub** and M. Pachter, "Cooperative Pursuit-Evasion Games With a Flat Sphere Condition," in IEEE Open Journal of Control Systems, vol. 4, pp. 360-372, 2025, doi: 10.1109/OJCSYS.2025.3604640.
- [5] Satyanarayana G. Manyam, D. W. Casbeer, A. Von Moll and **Isaac E. Weintraub**, "Dubins Path With Terminal Range and Field-of-View Constraints," in IEEE Control Systems Letters, vol. 9, pp. 1940-1945, 2025, doi: 10.1109/LCSYS.2025.3582622.
- [6] Abhishek Kashyap, Animesh Chakravarthy, Kamesh Subbarao, David Casbeer, **Isaac Weintraub**, and Brandon Hincey, "Macroscopic-Model-Based Learning for Detection of Cyber-Attacks in UAV Swarms", Journal of Aerospace Information Systems 2025 22:10, 867-889

- [7] Dejan Milutinović, Alexander Von Moll, **Isaac E. Weintraub**, and David Casbeer. "Stochastic Optimal Avoidance of Multiple Engagement Zones" *Journal of Aerospace Information Systems* 2025 22:6, 425-432
- [8] **Isaac E. Weintraub**, Alexander Von Moll, David W. Casbeer, and Satyanarayana G. Manyam, "Virtual Target Selection for a Multiple-Pursuer–Multiple-Evader Scenario", *Journal of Aerospace Information Systems* 2025 22:5, 391-400
- [9] D. Scott, Satyanarayana G. Manyam, **Isaac E. Weintraub**, David W. Casbeer and Mrinal. Kumar, "Noise Aware Path Planning and Power Management of Hybrid Fuel UAVs," in *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 8227-8238, 2025, doi: 10.1109/TASE.2024.3481998.
- [10] Alexander Von Moll and Isaac Weintraub, Basic Engagement Zones, *Journal of Aerospace Information Systems* 2024 21:10, 885-891
- [11] Pachter, M., Weintraub, I.E. The Synthesis of Optimal Control Laws Using Isaacs' Method for the Solution of Differential Games. *J Optim Theory Appl* 202, 1137–1157 (2024).  
<https://doi.org/10.1007/s10957-024-02490-7>
- [12] P. Kumar Ranjan, A. Sinha, Y. Cao, D. Casbeer and I. Weintraub, "Relational Maneuvering of Leader-Follower Unmanned Aerial Vehicles for Flexible Formation," in *IEEE Transactions on Cybernetics*, vol. 54, no. 10, pp. 5598-5609, Oct. 2024, doi: 10.1109/TCYB.2024.3435029.
- [13] Patrick Dillon, Michael Zollars, **Isaac Weintraub**, Alexander Von Moll, "Optimal Strategies for Avoidance of Multiple Weapon Engagement Zones", in *Journal of Aerospace Information Systems* 1.16 (2023), doi: 10.2514/1.1011224.
- [14] **Isaac Weintraub**, Alexander Von Moll, Eloy Garcia, David W Casbeer, and Meir Pachter, "Surveillance of a Faster Fixed-Course Target", in *IEEE Transactions on Aerospace and Electronic Systems*, vol. 59, no. 4, pp. 4147-4158, Aug. 2023, doi: 10.1109/TAES.2023.3237129.
- [15] Camilla Tabasso, Calvin ' Kielas-Jensen, Venanzio Cichella, Satyanarayana Manyam, David W Casbeer, **Isaac Weintraub**, "Continuous Monitoring of a Path-Constrained Moving Target by Multiple Unmanned Aerial Vehicles", in *Journal of Guidance, Control, and Dynamics* 45.4, pp 704–703 (2022), doi: 10.2514/1.G006043
- [16] Praveen Kumar Ranjan, Abhinav Sinha, Yongcan Cao, Dzung Tran, David Casbeer, and **Isaac Weintraub**, "Energy-Efficient Ring Formation Control with Constrained Inputs" in *Journal of Guidance, Control, and Dynamics*, vol. 46. Number 7, pp 1397-1407, doi: 10.2514/1.G007057
- [17] Dzung Tran, David Casbeer, Eloy Garcia, **Isaac Weintraub**, Dejan Milutinovic, "Ring Formation Maneuver: Double Integrator Kinematics with Input Saturation," in *Journal of Guidance, Control and Dynamics* (2021), doi: 10.2514/1.G005925.
- [18] Satyanarayana G Manyam, David W Casbeer, Swaroop Darbha, **Isaac Weintraub**, Krishna Kalyanam, "Path Planning and Energy Management of Hybrid Air Vehicles for Urban Air Mobility", in *IEEE Robotics and Automation Letters*, vol. 7, no. 4, pp. 10176-10183, Oct. 2022, doi: 10.1109/LRA.2022.3191810.
- [19] Calvin Kielas-Jensen, Venanzio Cichella, David Casbeer, Satyanarayana Gupta Manyam, **Isaac Weintraub**, "Persistent Monitoring by Multiple Unmanned Aerial Vehicles Using Bernstein Polynomials", in *Journal of Optimization Theory and Applications* (2021), doi: 10.1007/s10957-021-01921-z.
- [20] **Isaac Weintraub**, Alexander Von Moll, Eloy Garcia, and Meir Pachter, "Maximum Observation of a Target by a Slower Observer in Three Dimensions", *Journal of Guidance, Control, and Dynamics* 44.3 (2021), pp 646-653. doi:10.2514/1.G005619.
- [21] **Isaac Weintraub**, Eloy Garica, and Meir Pachter, "Optimal Guidance Strategy for the Defense of a Non-Manoeuvrable Target in 3-Dimensions", *IET Control Technology and Applications* 13.11 (2020), pp 1531-1538. doi: 10.1049/iet-cta.2019.0541.

- [22] Eloy Garcia, **Isaac Weintraub**, David Casbeer, "Adaptive Event-triggered Cooperative Control of Uncertain Networked Systems", IFAC-PapersOnLine 51.2 (2018), pp 82-87. doi:10.1016/j.ifacol.2018.07.092.
- [23] Michael W. Oppenheimer, **Isaac Weintraub**, David O. Sigthorsson, David B. Doman, "Control of a Minimally Actuated Biomimetic Vehicle Using Quarter-Cycle Wingbeat Modulation", Journal of Guidance, Control, and Dynamics 38.7 (2015), pp 1187-1196. doi: 10.2514/1.G000548.
- [24] **Isaac Weintraub**, David O. Sigthorsson, Michael W. Oppenheimer, David B. Doman, "Implementation of Split-Cycle Control for Micro Aerial Vehicles", Advances in Intelligent Systems and Computing 274:859-876, 2014. doi:10.1007/978-3-319-05582-4\_76.
- [25] David O. Sigthorsson, **Isaac Weintraub**, and Christopher Smithson, "Structural Technology Evaluation and Analysis Program (STEAP). Delivery Order 0035: Dynamics and Control and Computational Design of Flapping Wing Micro Air Vehicles", Tech report. DTIC Document ADA582345, 2012.
- [26] Asha J. Hall, Richard A. Roberts, **Isaac Weintraub**, Jaret C. Riddick, "Flapping Wing Technology for Micro Air Vehicles Incorporating a Lead Zirconate Titanate (PZT) Bimorph Actuator", Tech Report. DTIC Document (ADA564753), 2012.
- [27] Diane Evans and **Isaac Weintraub**, "Conditional Probability and the 50:50 Lifeline on Who Wants to Be a Millionaire?", Journal of Mathematics and Computer Education, 43.1 (2009), pp69-80.

## Conference

- [1] Rajnikant Sharma, Abhinav Sinha and **Isaac E. Weintraub**. "Learning-Augmented Adaptive Guidance for Target Interception," AIAA 2026-1202. AIAA SCITECH 2026 Forum. January 2026.
- [2] Alexander W. Denton, Michael P. Riddick, **Isaac E. Weintraub** and Donald L. Kunz. "Deep Neural Network as 5-D Image for Threat Avoidance," AIAA 2026-2194. AIAA SCITECH Forum. January 2026.
- [3] Qiang Le and **Isaac E. Weintraub**. "Basic Engagement Zone Avoidance Using Pseudospectral Methods," AIAA 2026-1200. AIAA SCITECH 2026 Forum. January 2026.
- [4] Reyshwanth Ganeshan, Sivakumar Rathinam, Swaroop Darbha, Satyanarayana G Manyam, **Isaac E. Weintraub** and David Casbeer. "Bi-Objective Dynamic Zone Navigation," AIAA 2026-1201. AIAA SCITECH 2026 Forum. January 2026.
- [5] Alexander L. Von Moll and **Isaac E. Weintraub**. "Reactive Vehicle Guidance Using Dynamic Maneuvering Cue," AIAA 2026-1199. AIAA SCITECH 2026 Forum. January 2026.
- [6] Grant Stagg, Cammy K. Peterson and **Isaac Weintraub**. "Probabilistic Weapon Engagement Zones for a Turn Constrained Pursuer," AIAA 2026-2195. AIAA SCITECH 2026 Forum. January 2026.
- [7] Alexander W. Denton, Donald Kunz, **Isaac E. Weintraub** and Alexander Von Moll, "The Basic Engagement Zone in 3-D," NAECON 2025 - IEEE National Aerospace and Electronics Conference, Dayton, OH, USA, 2025, pp. 1-4, doi: 10.1109/NAECON65708.2025.11235322.
- [8] Satyanarayana G. Manyam, D. W. Casbeer, A. Von Moll and **I. Weintraub**, "Shortest Dubins Path to a Moving Circle with Free Final Heading," 2025 American Control Conference (ACC), Denver, CO, USA, 2025, pp. 5046-5051, doi: 10.23919/ACC63710.2025.11107508.
- [9] **I. E. Weintraub** et al., "Min-Time Escape of a Dubins Car from a Polygon," 2025 American Control Conference (ACC), Denver, CO, USA, 2025, pp. 382-388, doi: 10.23919/ACC63710.2025.11107564.
- [10] A. Von Moll, D. Milutinović, **I. Weintraub** and D. W. Casbeer, "One-vs-one Threat-Aware Weaponing with Basic Engagement Zones," 2025 International Conference on Unmanned Aircraft Systems (ICUAS), Charlotte, NC, USA, 2025, pp. 1138-1145, doi: 10.1109/ICUAS65942.2025.11007913.
- [11] T. Chapman, **I. E. Weintraub**, A. Von Moll and E. Garcia, "Engagement Zones for a Turn Constrained Pursuer," 2025 International Conference on Unmanned Aircraft Systems (ICUAS), Charlotte, NC, USA, 2025, pp. 763-768, doi: 10.1109/ICUAS65942.2025.11007894.

- [12] Dejan Milutinović, Alexander L. Von Moll, **Isaac E. Weintraub** and David Casbeer. "Policy for Optimal Avoidance of Multiple Engagement Zones," AIAA 2025-1544. AIAA SCITECH Forum. January 2025.
- [13] Braulio Mora, Animesh Chakravarthy, Alexander Von Moll, **Isaac E. Weintraub** and David Casbeer. "Influence of Malicious Agents in a Team Seeking to Contain an Evader," AIAA 2025-1348. AIAA SCITECH. January 2025.
- [14] **Isaac E. Weintraub**, Alexander L. Von Moll, Alexander Hansen and Michael D. Zollars. "An Optimal Strategy for Off-Board Proximal Sensing of a Target: Part 1," AIAA SCITECH Forum. 2025.
- [15] Alexander Hansen, Michael D. Zollars, **Isaac E. Weintraub**, and Alexander L. Von Moll. "An Optimal Strategy for Off-Board Proximal Sensing of a Target: Part 2," AIAA 2025-1350. AIAA SCITECH. 2025.
- [16] Artur Wolek, David Casbeer, **Isaac E. Weintraub** and Alexander Von Moll. "Maximum Kinetic Energy Paths for a Decaying-Speed Dubins Vehicle," AIAA 2025-1351. AIAA SCITECH Forum. January 2025.
- [17] D. Milutinović, A. V. Moll, Satyanarayana G. Manyam, D. W. Casbeer, **I. E. Weintraub** and M. Pachter, "Pursuit-Evasion on a Sphere and When It Can Be Considered Flat," 2024 IEEE 63rd Conference on Decision and Control (CDC), Milan, Italy, 2024, pp. 8270-8275, doi: 10.1109/CDC56724.2024.10885887.
- [18] B. Mora, A. Von Moll, **I. Weintraub**, D. W. Casbeer and A. Chakravarthy, "Escape from an Orbiting Pursuer with a Nonzero Capture Radius," NAECON 2024 - IEEE National Aerospace and Electronics Conference, Dayton, OH, USA, 2024, pp. 200-205, doi: 10.1109/NAECON61878.2024.10670675.
- [19] D. Scott, Satyanarayana G. Manyam, D. W. Casbeer, M. Kumar and **I. E. Weintraub**, "Multi Agent Pathfinding for Noise Restricted Hybrid Fuel Unmanned Aerial Vehicles," 2024 American Control Conference (ACC), Toronto, ON, Canada, 2024, pp. 833-838, doi: 10.23919/ACC60939.2024.10644872.
- [20] **Isaac E. Weintraub**, A. Von Moll and M. N. Pachter, "Minimum Time Escape from a Circular Region of a Dubins Car," NAECON 2024 - IEEE National Aerospace and Electronics Conference, Dayton, OH, USA, 2024, pp. 34-37, doi: 10.1109/NAECON61878.2024.10670348.
- [21] Carl Gotwald, Michael D. Zollars, Jonah A. Reeger and **Isaac Weintraub**. "A Leader-Follower Control Strategy Built and Refined using Relational Maneuver Primitives for Approximating Optimal Trajectories in Real-Time," AIAA 2024-2760. AIAA SCITECH 2024 Forum. January 2024.
- [22] Abhishek Kashyap, Animesh Chakravarthy, Kamesh Subbarao, David Casbeer, **Isaac Weintraub** and Brandon Hency. "Modeling and Detection of Cyber-Attacks in UAV Swarms using a 2D-LWR Model and Gaussian Processes," AIAA 2024-0529. AIAA SCITECH 2024 Forum. January 2024.
- [23] Alexander L. Von Moll, David Casbeer, **Isaac Weintraub** and Meir Pachter. "Pure Pursuit of a Target on a Circular Trajectory," AIAA 2024-0956. AIAA SCITECH 2024 Forum. January 2024.
- [24] **Isaac Weintraub**, Alexander L. Von Moll, David Casbeer and Satyanarayana G. Manyam. "Virtual Target Selection for a Multiple-Pursuer Multiple-Evader Scenario," AIAA 2024-0123. AIAA SCITECH 2024 Forum. January 2024.
- [25] **Isaac Weintraub**, Alexander Von Moll, Meir Pachter, "Range-Limited Pursuit-Evasion," NAECON 2023 - IEEE National Aerospace and Electronics Conference, Dayton, OH, USA, 2023, pp. 28-35, doi: 10.1109/NAECON58068.2023.10365808.
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## Technical Lectures

- [1] D. Milutinovic, A. Von Moll, **I. Weintraub**, D. Casbeer, “Stochastic Risk-Aware Path Planning Around Multiple Threats,” at AIAA DEFENSE Forum, Laurel, MD, April 2024.
- [2] **I. Weintraub**, A. Wolek, A. Von Moll, D. Casbeer, S. Manyam, “Deterministic Risk-Aware Path-Planning Around Multiple Threats,” at AIAA DEFENSE Forum, Laurel, MD, April 2024.
- [3] A. Von Moll, **I. Weintraub**, “Digitally Enhanced Aim-Point for Capture of Mobile Targets,” at AIAA DEFENSE Forum, Laurel, MD, April 2024.
- [4] C. Gotwald, M. Zollars, **I. Weintraub**, J. Reeger, “Aircraft Control Strategy Using Relational Maneuvering Primitives,” at AIAA DEFENSE Forum, Laurel, MD, April 2024.
- [5] **I. Weintraub**. “Risk-Aware Autonomy,” at Coalition Warfare Program Team (OUSD), Alexandria, VA, 17 Jan 2024.

[6] **I. Weintraub**, Alexander Von Moll, Nicholas Hanlon, “Automatic Engagement Zone Avoidance: Theory and Practice”, at AIAA DEFENSE Forum, Laurel, MD, April 2023.

**White Papers**

[1] A. Von Moll, **I. Weintraub**, “Multi-Agent Risk-aware Teaming under Uncertainty (MARTY),” Air Force Research Laboratory, 2024.

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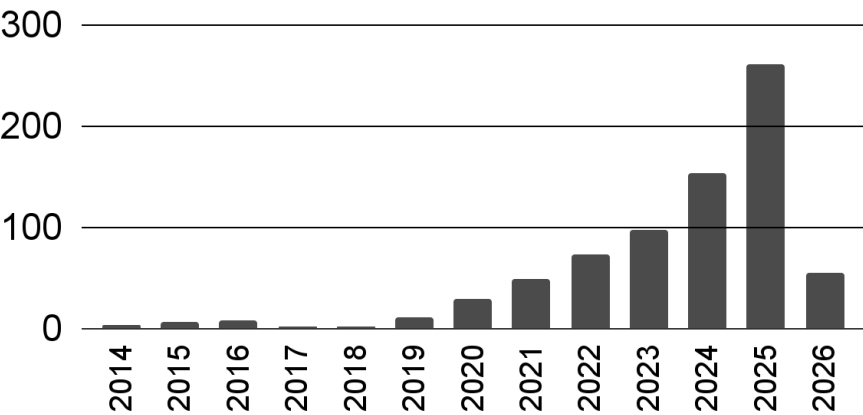
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[1] “Mission Driven Autonomy,” presented at Rose-Hulman Institute of Technology, Terre Haute, IN, 2/12/2025

[2] “Mission Aware Autonomy,” presented at the Aerospace Control and Guidance Systems Committee Meeting, Asheville, NC, 11/2024

[3] “Professional Graphics for Publication,” presented at Air Force Institute of Technology, 9/19/2023

[4] “An Optimal Engagement Zone Avoidance Scenario in 2-D,” presented at Air Force Institute of Technology, 4/24/2023

[5] “Professional Graphics for Publication,” presented at Air Force Institute of Technology, 9/21/2022

[6] “Optimal Control Direct Methods,” presented at Air Force Institute of Technology, 5/23/2022

[7] “Getting your Hands Dirty”, presented at Wright-Brother’s Institute, 08/04/2017

**Professional Organizations**

|                                                                  |                   |
|------------------------------------------------------------------|-------------------|
| <b>American Institute of Aeronautics and Astronautics (AIAA)</b> | Member Since 2011 |
| Associate Fellow, Lifetime Member                                |                   |
| Member: Intelligent Systems Technical Committee                  | 2021 - present    |
| ISTC Executive Secretary                                         | 2025 - present    |

|                                                                          |                |
|--------------------------------------------------------------------------|----------------|
| ISTC Awards Subcommittee                                                 | 2022 - 2024    |
| ISTC Technical Discipline Chair                                          | 2023 - 2024    |
| Member: Advanced Air Mobility Task Force                                 | 2022           |
| Outstanding Peer Reviewer of JGCD (Letter from Ping Lu, Editor-in-Chief) | 2020 - present |
| AIAA SciTech, Session Chair                                              | 2022 - present |
| AIAA DEFENSE Forum, Session Chair                                        | 2024 - 2025    |
| AIAA GNC TC Member                                                       | 2026 - present |

## **Institute of Electrical and Electronics Engineers (IEEE)**

Member Since 2009

Senior Member

Member: Controls Systems Society, Robotics and Automation Society, Young Professionals

Tutorial Session Co-organizer, "Introduction to Pursuit Evasion Differential Games" 2020

IEEE Conference on Control Technology and Applications. Session Chair 2020

IEEE Student Conference Chair, University Texas at Arlington 2009

IEEE Student Publicity Chair, University of Texas at Arlington 2008

## **Academic Organizations**

Tau Beta Pi (Member), Eta Kappa Nu (Member), Pi Mu Epsilon (Member), Order of the Engineer (Member)

## **Interests and Extracurricular Activities**

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Running Races</b>    | 2-Marathons, 9-Half-marathons, 8-10Ks, 4-5Ks, 4-Relays                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Ballroom Dance</b>   | Arthur Murray bronze level certification                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Tae-Kwon-Do</b>      | 4th degree black belt (instructor); taught women's self-defense classes; won in sparring at numerous local competitions in sparring and patterns; won 3rd place at three international tournaments for sparring                                                                                                                                                                                                                                                        |
| <b>Building Comp</b>    | Two-time "Boneyard Build-off" Participant Won 2nd place in 2018                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>FIRST Robotics</b>   | Volunteer Coach, Bonds #5811 (2021-2022 Season)<br>Volunteer Coach, Dayton Regional STEM School<br>First Tech Challenge Teams #5345 and #4537<br>Student Participant, Toledo Technology Academy<br>First Robotics Competition, Team #279<br>2005 National Championship - Finalist; 2005 Buckeye Regional - GM Industrial Design Award, Regional Winners, and Safety Award Winner; 2004 Great Lakes Regional - Regional Winner; 2004 Buckeye Regional - Regional Winner |
| <b>Antique Vehicles</b> | Frame off Restoration of a 1966 Chevy C10<br>Restored 1976 Honda CB750; 3rd place in a motorcycle restoration competition                                                                                                                                                                                                                                                                                                                                              |
| <b>Home Brewing</b>     | I have been an avid homebrewer for many years. I make wines, meads, and beer                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Piano</b>            | I play the piano, albeit I'm a bit rusty                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Youtube</b>          | I have begun making instructional videos to assist others in typesetting and scientific writing in the Typst programming language.                                                                                                                                                                                                                                                                                                                                     |

## **References**

- References available on request
- Eligible for Security Clearance and Polygraph
- Active TS/SC//TK/HCS/SI